



OPEN The impact of sex/gender-specific funding and editorial policies on biomedical research outcomes: a cross-national analysis (2000–2021)

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Reflecting sex and gender characteristics in biomedical research is critical to improving health outcomes and reducing adverse effects from medical treatments. This study investigates the impact of sex/gender-specific funding policies and journal editorial standards on the integration of sex/gender analysis in biomedical research publications. Using data from the United States, Canada, the United Kingdom, and other countries between 2000 and 2021, we assessed how these policies influenced research output in the fields of medicine and life sciences. Our findings show that countries with progressive funding policies and journals promoting sex/gender-based reporting have significantly improved research quality and publication rates. This highlights the importance of coordinated policy efforts and editorial practices in advancing integrated sex/gender research. We recommend continued global efforts from policymakers, funding bodies, and journals to embed sex/gender perspectives in scientific inquiry, ensuring more effective and equitable biomedical advancements.

Keywords Biomedical research, Sex/gender differences, Integrated-sex/gender, Funding, Journal editorial policy, Policy effects

The importance of considering sex and gender characteristics in biomedical research has been increasingly recognized over the past two decades^{1–8}. Sex refers to biological differences between males and females, such as chromosomes, hormone levels, and reproductive anatomy, while gender refers to the socially constructed roles, behaviors, and identities that society considers appropriate for men, women, and non-binary individuals. Both sex and gender play a crucial role in health outcomes, influencing disease risk, progression, and treatment effectiveness^{9–14}. For example, sex-based biological differences, such as variations in immune response between males and females, can affect susceptibility to diseases and reactions to vaccines^{2,5,8,14,15}. Gender, on the other hand, impacts health behaviors, such as how individuals seek medical care or adhere to treatment protocols¹⁶, which in turn affects the course of many conditions, including mental health disorders^{17–20} and cardiovascular^{10,16,21} disease. Line this end, Physiological, genetic, and environmental differences between the sexes contribute to sex differences in a variety of conditions. However, scientific research has traditionally been single-sex or male-centered, assuming that all sexes are biologically similar. This has led to a significant gap in fundamental scientific understanding of sex-specific characteristics and mechanisms. The focus on one sex has caused issues in drug development and in the prevention, diagnosis, treatment, and management of diseases, leading to adverse drug reactions and decreased efficacy across broad domains^{10–14,22}. These problems not only seriously damage people's health and quality of life, but also pose a major challenge to the development of the entire social welfare system and the pharmaceutical industry^{6,8,23}.

Furthermore, as biological sex differences have been revealed by sex hormones, sex chromosomes, and socio-cultural environments, new sex differences are expanding our knowledge of sex difference in various biomedical fields, including the immune system^{2,5,8,14}, cancer^{24–26}, mental health^{7,18,19,27–32}, and cardiovascular diseases^{16,21,33,34}. In response to this need, numerous international organizations have introduced policies to promote integrated sex/gender research in the biomedical sciences. Organizations such as the National Institutes of Health (NIH) in the United States³⁵, the Canadian Institutes of Health Research (CHIR) in Canada^{36,37}, and the European Commission in Europe^{38,39} are increasingly rewarding excellence in integrated-sex/gender research in the biomedical sciences. For example, the NIH has emphasized the importance of integrated-sex/gender

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research, and its funding reflects this. To promote integrated-sex/gender research, the NIH recently released the “NIH Inclusion Policy,” which requires researchers to include sex/gender-specific experimental subjects in their protocols. Canada’s CHIR also actively supports integrated-sex/gender research. CHIR encourages integrated-sex/gender research to advance our understanding of gender-specific health issues, and some of its funding programs prioritize integrated-sex/gender research and provide opportunities to share and apply the results of integrated-sex/gender research. UK Research and Innovation (UKRI), one of the UK’s research funding agencies, has a policy to promote research with a gender dimension⁴⁰. UKRI emphasizes the importance of research that considers sex/gender differences and encourages researchers to actively incorporate gender dimensions. In this regard, UKRI aims to use integrated-sex/gender research to investigate sex/gender-specific health issues and to explore ways to reduce discrimination based on gender. UKRI also considers sex/gender differences in its research funding to help researchers collect sex/gender-disaggregated data, analyze results, and make policy recommendations. The efforts of these organizations recognize the importance of integrated-sex/gender research in the biomedical sciences and strengthen research support policies to promote gender-responsive research and spread awareness^{36,41,42}.

Highly impactful scientific journals such as *Nature*, *Cell*, and *The Lancet* have also recognized the quality and reliability of integrated-sex/gender research in biomedical research and have recommended it^{22,43–46}. These prestigious journals encourage researchers to report sex/gender characteristics in their methodologies, analyses, and discussion sections when publishing their work^{46,47}. For example, *Nature* asks authors to describe their methodology and results in detail. *Cell* and *Lancet* also actively support research that takes into consideration sex/gender characteristics. This shift in editorial standards reflects a broader acknowledgment within the scientific community of the value of integrated sex/gender research for producing more reliable and applicable findings.

While the importance of integrated-sex/gender research is increasingly recognized, there is still a lack of research and awareness^{1,6,20,48}. There is a need to explore ways to more efficiently disseminate integrated-sex/gender research. To address this gap, this study examines the impact of sex/gender-specific funding policies and journal editorial standards on research publication outcomes across multiple countries, including the United States, Canada, and the United Kingdom. By analyzing data from 2000 to 2021, this study aims to assess how progressive funding and editorial policies have influenced the volume of sex/gender-integrated biomedical research. The findings will contribute to a better understanding of the effectiveness of these policies and provide recommendations for fostering further integration of sex/gender considerations in global biomedical research.

Materials and methods

Data source

We conducted an extensive literature review modified the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) protocol. This review utilized the Dimensions AI, an advanced data and analytics platform tailored for researchers, academic institutions, and funding agencies.

Study selection and screening

Our review focused on a range of biomedical fields—including Biomedical and Clinical Sciences, Health Sciences, Psychology, Biological Sciences, Public Health, Biological Psychology, Health Service, Clinical and Health Psychology, and Clinical Science—where research on sex/gender trait reflection is considered important^{1,42}. In line with our study objectives, we employed specific search terms associated with sex/gender trait reflection, querying these terms within the “Title” and “Abstract” fields of the Dimensions AI database to ensure a focused retrieval of relevant academic articles⁴⁹.

A total of 5,103,170 articles (4,410,763 grants) were screened from the Dimensions AI database using the following search terms in the “Title” and “Abstract” fields:

“sex factor*” OR “sex characteristic*” OR “sex difference*” OR “gender factor*” OR “gender characteristic*” OR “gender difference*” OR “sex specific*” OR “gender specific*”

NOT (“sexual partner*” OR “sexual selection*” OR “sexual behavior*” OR “sexual behavior*”).

Out of these 5,103,170 articles (4,410,763 grants), 324,639 (21,250 grants) studies were included based on their direct reflection of sex/gender characteristics in biomedical research—including Biomedical and Clinical Sciences, Health Sciences, Psychology, Biological Sciences, Public Health, Biological Psychology, Health Service, Clinical and Health Psychology, and Clinical Science—where research on sex/gender trait reflection is considered important^{1,42}. The inclusion criteria required that the articles either employed sex/gender-disaggregated data or examined sex/gender differences in health outcomes. Articles focusing on sexual behavior or selection were excluded to maintain relevance to the study’s objectives.

Data collection was performed for the time period of 2000 to 2021, to capture trends before and after the implementation of key sex/gender-specific policies in the United States, Canada, the United Kingdom, and other countries. The Dimensions AI platform was accessed on March 3, 2023, to retrieve the final dataset.

Statistics

Research papers or grant reflecting sex/gender characteristics was measured by calculating the ratio of the total number of papers (and/or grant) published to the number of papers (or grant) containing the keyword as a percentage.

To evaluate the impact of sex/gender-specific funding policies and journal editorial standards, we employed several statistical methods using Graph pad 10. In the resulting graph, each method and figure is noted:

1. **Linear Regression Analysis:** We used simple linear regression to assess trends over time, focusing on the annual increase or decrease in research publications and funding across each country. This analysis allowed us to identify overall trends in sex/gender-i research output.
2. **Interrupted Time-Series (ITS) Analysis:** To assess the effectiveness of policy interventions, such as the NIH's SABV policy (2016, United States), CIHR's GBA + framework (2017, Canada), and UKRI's gender dimension policy (2018, United Kingdom), we conducted Interrupted Time-Series (ITS) analysis. ITS is a robust method for evaluating the impact of policy changes over time by comparing data before and after the implementation of these policies.
 - **Modeling Approach:** For each country, we constructed ITS models to examine the immediate effect of the policy (interruption) and the change in trends over time (interaction). The ITS model included variables to represent the time before the intervention, the immediate impact of the intervention (interruption), and the trend after the intervention (interaction term).
 - **Outcome Variables:** The key outcome variables were the number of research grants and the number of publications that included sex/gender-sensitive terms, measured annually from 2000 to 2021.
3. **Pearson Correlation:** Pearson correlation coefficients were calculated to evaluate the strength of the relationship between the number of grants awarded and the number of publications in each country. This analysis allowed us to measure the extent to which research funding correlates with publication output in sex/gender-sensitive research.
4. **Unpaired T-Test:** We conducted unpaired t-tests to compare the average number of research grants and publications before and after the implementation of sex/gender-sensitive policies in each country. This analysis helped identify significant differences in research outputs following policy changes.

Country-specific analysis

Our analysis focused on five countries: the United States, Canada, the United Kingdom, Japan, and China. These countries were selected because they represent different stages of policy development and implementation for sex/gender-sensitive research.

- **United States:** The SABV policy was introduced by the NIH in 2016, requiring the inclusion of both male and female subjects in preclinical and clinical research.
- **Canada:** The GBA + framework, adopted by CIHR in 2017, promotes the integration of sex and gender analysis in health research.
- **United Kingdom:** The UKRI policy on sex/gender considerations in research was implemented in 2018, encouraging researchers to consider gender-specific risks and outcomes.
- **Japan and China (Korea, only papers):** These countries do not yet have formal sex/gender-sensitive policies in place, providing a comparison group for examining the impact of policy versus non-policy environments.

Field-specific analysis

We also analyzed how sex/gender-integrated research varied across different disciplines. To identify fields where sex/gender integration was most prevalent, we focused on four key research categories:

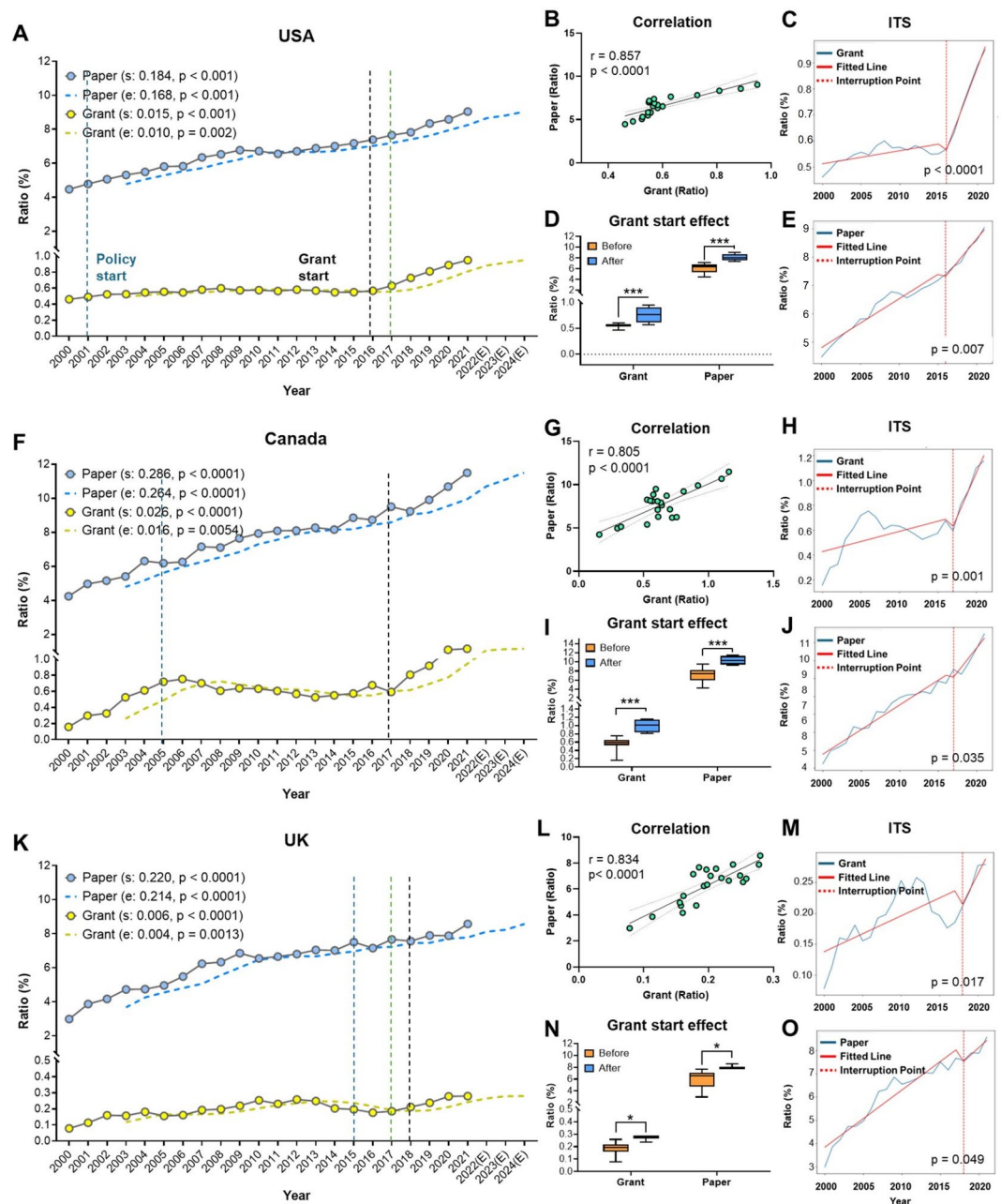
1. Psychology
2. Biological sciences
3. Biomedical and Clinical sciences
4. Health Sciences

By calculating the annual trends in research grants and publications for each field, we were able to identify which fields have shown the greatest growth in sex/gender-sensitive research. For example, psychology showed the highest level of integrated sex/gender research support and publication output, driven largely by the significant sex/gender disparities in mental health conditions.

Results

Patterns between integrated-sex/gender research funding rates and publication outcomes

To explore ways to more efficiently disseminate integrated-sex/gender research, this study examined and analyzed the number of research papers, funding amounts, and related policies in the biomedical field reported by country from 2000 to 2021 (Fig. 1, Supple 1). In the USA (Fig. 1A), Canada (Fig. 1F), and the UK (Fig. 1K), a positive trend was identified with significant increases in both the size of sex/gender-specific grants and corresponding research outputs. In the USA, the overall publication output displayed a statistically significant positive slope (0.184, $p < 0.0001$), with a strong Pearson correlation ($r = 0.857$, $p < 0.0001$) between funding and publication outcomes. The data also projected that both research funding and outputs are expected to increase by 2024, continuing the observed positive trend (expected slope = 0.168, $p < 0.0001$) (Fig. 1A-E). In Canada (Fig. 1F), a positive trend was observed, with significant increases in both the size of sex/gender-specific grants and corresponding research outputs. The overall publication output in Canada showed a statistically significant positive slope of 0.286 ($p < 0.0001$), and there was a strong Pearson correlation of $r = 0.805$ ($p < 0.001$) between funding and publication outcomes. Projections indicate that both research funding and publication outputs are expected to continue rising through 2024, with the expected slope for publications being 0.264 ($p < 0.0001$) (Fig. 1F-J). In the UK (Fig. 1K), a similar positive trend was identified, with notable increases in sex/gender-specific grants and research outputs. The publication output in the UK demonstrated a statistically significant positive



slope of 0.220 ($p < 0.0001$), with a strong Pearson correlation of $r = 0.834$ ($p < 0.0001$) between funding and publication outcomes. The data further predict that both funding and research outputs will continue to rise by 2024, with the projected publication slope being 0.214 ($p < 0.0001$) (Fig. 1K–O). By contrast, Japan and China (Supple 1) showed weaker correlations between funding and research outcomes.

Furthermore, we found that after the strengthening of integrated-sex/gender research funding policies in each country, not only the amount of funding but also research performance increased significantly. In the USA (Fig. 1D), not only the amount of funding (before; 0.550 ± 0.009 , after; 0.762 ± 0.060 , unpaired $t = 5.544$, $p < 0.0001$) but also research performance (before; 6.094 ± 0.213 , after; 8.142 ± 0.255 , unpaired $t = 5.346$, $p < 0.0001$) increased significantly around 2016, when NIH was required to reflect SABV (sex as a biological variable)⁵⁰. Similarly, in Canada (Fig. 1I), not only the amount of funding (before; 0.560 ± 0.036 , after; 0.996 ± 0.081 , unpaired $t = 5.077$, $p < 0.0001$) but also research performance (before; 7.116 ± 0.355 , after; 10.329 ± 0.489 , unpaired $t = 4.038$, $p = 0.0006$) increased significantly around 2017, when CIHR was required to reflect GBA (gender-based analysis)⁵¹. And in the UK (Fig. 1N), not only the amount of funding (before; 0.189 ± 0.011 , after; 0.265 ± 0.014 , unpaired $t = 2.783$, $p = 0.0115$) but also research performance (before; 6.018 ± 0.319 , after; 8.119 ± 0.228 , unpaired $t = 2.554$, $p = 0.0189$) increased significantly around 2018, when UKRI promoted the gender dimension⁴⁰. This shows that funding policies have a direct and significant impact on research outputs, demonstrating the importance of sex/gender-inclusive funding policies to promote more sex/gender-inclusive research outputs.

To further confirm the effectiveness of the policy, we analyzed Interrupted Time-Series (ITS) before and after the year the research funding policy was implemented. ITS is a robust method for evaluating the

◀ **Fig. 1.** Trend of Integrated-sex/gender research grant and paper(A-E) In USA, (A) Overall pattern (Simple linear regression, Paper Slope (s); 0.184, $p < 0.0001$, Paper 3 Year expected trend (e); 0.168, $p < 0.0001$, Grant s; 0.015, $p < 0.0001$, Grant e; 0.010, $p = 0.0002$). (B) Correlation between grant and paper (Pearson $r = 0.857$, $p < 0.0001$). (C) USA Grant ITS; $r^2 = 0.967$, Time; Coefficient = 0.0050, $p < 0.001$, Interruption; Coefficient = -1.2085, $p < 0.001$, Interaction; Coefficient = 0.0738, $p < 0.0001$. (D) Comparison of Grant and paper ratio before and after the funding policy; Before Grant = 0.550 ± 0.009 , After grant = 0.762 ± 0.060 , Before Paper = 6.094 ± 0.213 , After Paper = 8.142 ± 0.255 (Unpaired t-test, Grant; $t = 5.544$, $p < 0.0001$, Paper; $t = 5.346$, $p < 0.0001$). (E) USA Paper ITS; $r^2 = 0.974$, Time; Coefficient = 0.1722, $p < 0.0001$, Interruption; Coefficient = -2.7712, $p = 0.009$, Interaction; Coefficient = 0.1581, $p = 0.007$. (F-J) In Canada, (F) Overall pattern (Simple linear regression, Paper s; 0.286, $p < 0.0001$, Paper e; 0.264, $p < 0.0001$, Grant s; 0.026, $p < 0.0001$, Grant e; 0.016, $p = 0.0054$). (G) Correlation between grant and paper (Pearson $r = 0.805$, $p < 0.001$). (H) Canada Grant ITS; $r^2 = 0.874$, Time; Coefficient = 0.0089, $p < 0.001$, Interruption; Coefficient = -0.1457, $p = -0.1457$, Interaction; Coefficient = 0.0035, $p = 0.001$. (I) Comparison of Grant and paper ratio before and after the funding policy; Before Grant = 0.560 ± 0.036 , After grant = 0.996 ± 0.081 , Before Paper = 7.116 ± 0.355 , After Paper = 10.329 ± 0.489 (Unpaired t-test Grant; $t = 5.077$, $p < 0.0001$, Paper; $t = 4.038$, $p = 0.0006$). (J) Canada Paper ITS; $r^2 = 0.929$, Time; Coefficient = 0.0376, $p < 0.0001$, Interruption; Coefficient = -0.0056, $p = 0.842$, Interaction; Coefficient = 0.0025, $p = 0.035$. (K-O) In UK, (K) Overall pattern (Simple linear regression, Paper s; 0.220, $p < 0.0001$, Paper e; 0.214, $p < 0.0001$, Grant s; 0.006, $p < 0.0001$, Grant e; 0.004, $p = 0.0013$). (L) Correlation between grant and paper (Pearson $r = 0.834$, $p < 0.0001$). (M) UK Grant ITS; $r^2 = 0.879$, Time; Coefficient = 0.0089, $p = 0.003$, Interruption; Coefficient = -0.0256, $p = 0.463$, Interaction; Coefficient = -0.0012, $p = 0.017$. (N) Comparison of Grant and paper ratio before and after the funding policy; Before Grant = 0.189 ± 0.011 , After grant = 0.265 ± 0.014 , Before Paper = 6.018 ± 0.319 , After Paper = 8.1119 ± 0.228 (Unpaired t-test Grant; $t = 2.783$, $p = 0.0115$, Paper; $t = 2.554$, $p = 0.0189$). (O) UK Paper ITS; $r^2 = 0.905$, Time; Coefficient = 0.0412, $p < 0.0001$, Interruption; Coefficient = 0.0156, $p = 0.663$, Interaction; Coefficient = -0.0031, $p = 0.049$. s (slope), Paper (Blue), Grant (Yellow), GAB policy (blue dash), Grant policy (Black dash), Journal editorial (Green dash), Before (Orange), After (Blue), interruption point (Red dash). Observed data (Blue line), Fitted line (Red line) Mean $\pm$ SEM, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

impact of policy changes over time by comparing data before and after the implementation of these policies. The ITS model included variables to represent the time before the intervention, the immediate impact of the intervention (interruption), and the trend after the intervention (interaction term). In the USA, the policy event in 2016, when the National Institutes of Health (NIH) introduced the requirement to include sex as a biological variable (SABV), had an immediate and significant positive effect on both grants and papers. The ITS analysis showed a significant positive impact on grants, with an interaction coefficient of 0.0738 ($p < 0.0001$) and an immediate effect with an interruption coefficient of -1.2085 ($p < 0.001$) (Fig. 1C). For publications, the effect was similarly strong, with an interaction coefficient of 0.1581 ($p = 0.007$) and an interruption coefficient of -2.7712 ($p = 0.009$) (Fig. 1E). This demonstrates that the 2016 NIH policy not only increased research funding but also led to a notable rise in sex/gender-sensitive research outputs. In Canada, the 2017 policy event, which saw the implementation of the Gender-Based Analysis Plus (GBA+), had no significant immediate impact on grants or publications. However, in the long term, the ITS analysis revealed a significant positive effect on grants, with an interaction coefficient of 0.0035 ($p = 0.001$) (Fig. 1H), and on papers, with an interaction coefficient of 0.0025 ($p = 0.035$) (Fig. 1J). Although there was no immediate shift following the 2017 policy, these findings suggest that over time, the GBA+ initiative has contributed to sustained increases in both funding and publication output. In the UK, the policy event in 2018, when UK Research and Innovation (UKRI) promoted the inclusion of the gender dimension in research, did not show a significant immediate impact on either grants or papers. The ITS analysis revealed that the interruption coefficients for both grants and papers were not statistically significant, with values of -0.0256 ($p = 0.463$) for grants and 0.0156 ($p = 0.663$) for papers (Fig. 1M and O). However, the interaction term for papers was marginally significant, with a coefficient of -0.0031 ($p = 0.049$), indicating that more time and data may be required to fully assess the long-term impact of the 2018 policy change. These ITS findings provide robust evidence that while the USA and Canada have seen clear positive long-term impacts from their respective policy changes, the UK may need more time and further data collection to assess the full effect of its 2018 gender-related research funding policies.

The EU has also emphasized the importance of sex/gender-inclusive research, initiating efforts to promote integrated-sex/gender research through Horizon 2020 in 2013, and fully supporting such research through Horizon Europe in 2014. Previous studies have shown that funding increased around this time, and the results have been steadily increasing^{36,39}.

Trends in integrated-sex/gender research by category in the USA and Canada

To identify specific areas of research that are directly affected by the consideration of gender characteristics, we analyzed funding and outcomes in each of the top four research categories (Psychology, Biological sciences, Biomedical and clinical sciences, and Health). The results show that both the USA and Canada have increased funding in all fields (Fig. 2A&B), but with different patterns of papers (Fig. 2C&D). In Psychology, funding in the USA showed a significant increase, with a slope of 0.061 ($p < 0.0001$), accompanied by a similarly sharp rise in publication output, with a slope of 0.231 ($p < 0.0001$). In Canada, funding also grew, with a slope of 0.035 ($p = 0.0008$), while publication output increased even more steeply, with a slope of 0.415 ($p < 0.0001$). These findings suggest that Psychology has seen robust support in both countries, leading to consistent increases in research outcomes, likely driven by the growing focus on mental health and sex/gender disparities. In Biological

Category Classification

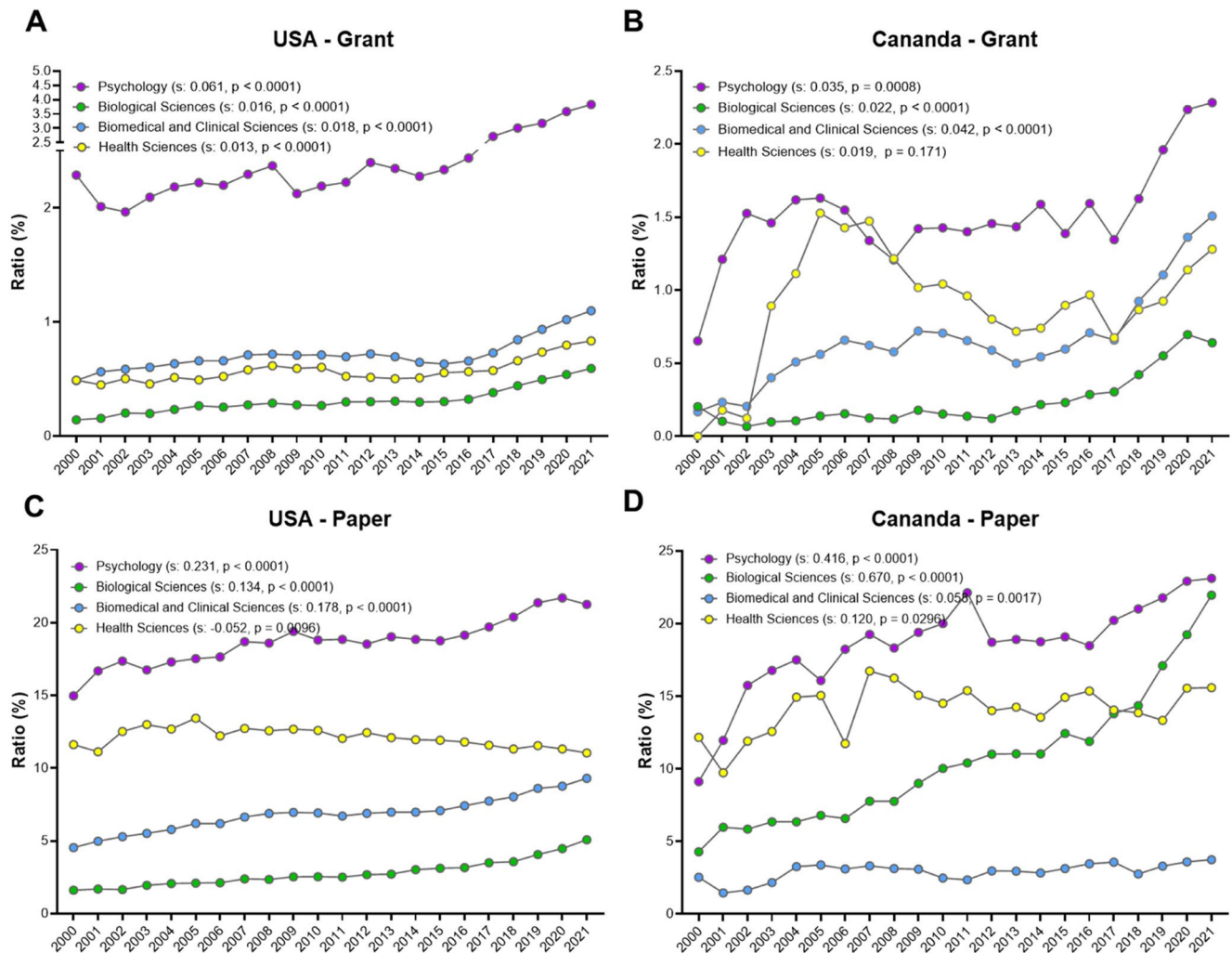


Fig. 2. Patterns by category of research reflecting sex/gender characteristics in the USA and Canada (A&C) In USA (A) Grant funding patterns (Simple linear regression, Psychology, s: 0.061, $p < 0.0001$, Biological science, s: 0.016, $p < 0.0001$, Biomedical and Clinical science: s: 0.018, $p < 0.0001$, Health science: s: 0.013), (C) Paper patterns (Simple linear regression, Psychology: s: 0.231, $p < 0.0001$, Biological science: s: 0.134, $p < 0.0001$, Biomedical and Clinical science: s: 0.178, $p < 0.0001$, Health science: s: -0.052, $p = 0.0096$). (B&D) In Canada (B) Grant funding patterns (Simple linear regression, Psychology: s: 0.035, $p = 0.0008$, Biological science: s: 0.022, $p < 0.0001$, Biomedical and Clinical science: s: 0.042, $p < 0.0001$, Health science: s: 0.019, $p = 0.171$), (D) Paper patterns (Simple linear regression, Psychology: s: 0.415, $p < 0.0001$, Biological science: s: 0.670, $p < 0.0001$, Biomedical and Clinical science: s: 0.058, $p = 0.0017$, Health science: s: 0.120, $p = 0.0296$). Slope(s), Psychology (Purple), Biological science (Green), Biomedical and Clinical science (Blue), Health science (Yellow).

Sciences, funding in the USA saw steady growth, with a slope of 0.016 ($p < 0.0001$), and publication output followed a similar pattern, with a slope of 0.134 ($p < 0.0001$). Canada experienced an increase in funding, with a slope of 0.022 ($p < 0.0001$), and an even more pronounced rise in publication output, with a slope of 0.670 ($p < 0.0001$). This trend in both countries highlights the increasing integration of sex/gender research in biological fields, where sex differences in biological mechanisms are becoming more prominent. In Biomedical and Clinical Sciences, funding in the USA showed positive growth, with a slope of 0.018 ($p < 0.0001$), while publication output increased, with a slope of 0.178 ($p < 0.0001$). In Canada, funding also saw an increase, with a slope of 0.042 ($p < 0.0001$), though the rise in publication output was more modest, with a slope of 0.058 ($p = 0.0017$). In Health Sciences, the USA experienced a steady increase in funding, with a slope of 0.013, but publication output actually declined, showing a negative slope of -0.052 ($p = 0.0096$). In contrast, Canada showed a slight positive trend in funding, with a slope of 0.019 ($p = 0.171$), and a moderate increase in publication output, with a slope of 0.120 ($p = 0.0296$). The disparity in the USA between increased funding and declining publication output in Health Sciences suggests potential inefficiencies in how gender considerations are being

incorporated or gaps in translating funding into meaningful research outcomes. Conversely, Canada's moderate rise in output reflects a more balanced growth between funding and research results. The correlations between research funding and research output were all positive, but not significant for health sciences (Supple 2). When analyzing funding and output rates by discipline in the UK (Supple 3), we found that funding was marginally positively sloped in all disciplines, and output was positively sloped in all but health sciences.

Impact of journal editorial policies

According to the pattern of research support and research performance in Japan (Supple1), we can observe that the scale of research funding is negatively sloped. However, the number of integrated-sex/gender research papers has been increasing. In China, there is no specific policy to support integrated-sex/gender research, and the number of publications has increased only marginally. In detail, we can observe a steady increase in performance since 2017.

In 2017, prominent journals in the biomedical field, such as Nature, Cell, and Lancet, emphasized the reliability and excellence of integrated-sex/gender research in their articles and editorial policies. To analyze the impact of these editorial policies on the performance of integrated-sex/gender research papers, we observed the status of published integrated-sex/gender research papers for about five years before and after the editorial policies. As a result (Fig. 3), we observed a significant increase in the performance of integrated-sex/gender research papers in Japan, China, and South Korea, which do not have integrated-sex/gender research funding policies (unpaired t-test). The USA showed a 1.2% increase in sex/gender-reflective papers from 2012 to 2016 (7.037 ± 0.117) to 2017–2021 (8.292 ± 0.252) ($t = 4.513$, $p = 0.002$). Canada also saw a notable increase, with publications rising from 8.426 ± 0.155 before the policies to $10.164 (\pm 0.413)$ afterward (unpaired t-test: $t = 3.945$, $p = 0.0043$). In the UK, sex/gender-reflective papers increased from 7.105 ± 0.115 to 7.918 ± 0.176 (unpaired t-test: $t = 3.870$, $p = 0.0047$), indicating a significant rise following the adoption of editorial policies. Japan, which has no dedicated sex/gender research funding policies, showed an increase from 4.17 ± 0.122 before the policies to 5.659 ± 0.242 after the editorial changes (unpaired t-test: $t = 4.584$, $p = 0.0018$). Similarly, China saw a moderate increase in the ratio of sex/gender-reflective publications, from 3.718 ± 0.147 to 4.292 ± 0.241 (unpaired t-test: $t = 2.739$, $p = 0.0255$). In South Korea, where there is also no specific funding for sex/gender research, the ratio of publications increases significantly from 4.010 ± 0.218 before the policies to 5.647 ± 0.271 after the editorial policies (unpaired t-test: $t = 4.713$, $p = 0.0015$) (Fig. 3). These findings illustrate that journal editorial policies have had a direct and significant impact on research performance in promoting sex/gender-sensitive research across all countries, even in regions without dedicated funding support.

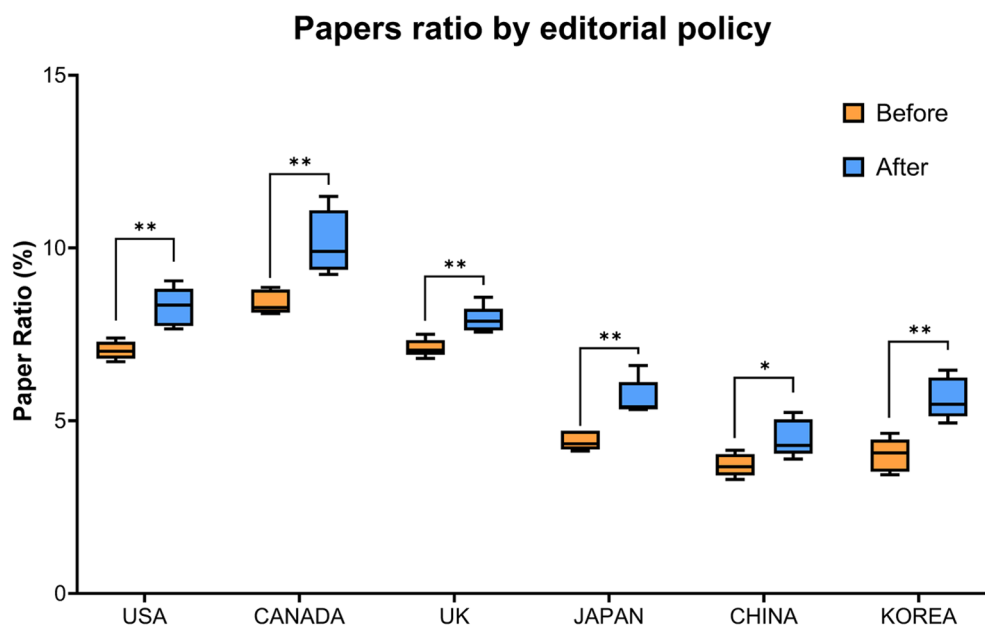


Fig. 3. Comparison of proportion of sex/gender-reflective papers before (2012–2016) and after (2017–2021) journal editorial policies. USA; Before = 7.037 ± 0.117 , After = 8.292 ± 0.252 (unpaired t test, $t = 4.513$, $p = 0.002$), CANADA; Before = 8.426 ± 0.155 , After = 10.164 ± 0.413 (unpaired t test, $t = 3.945$, $p = 0.0043$), UK; Before = 7.105 ± 0.115 , After = 7.918 ± 0.176 (unpaired t test, $t = 3.870$, $p = 0.0047$), Japan; Before = 4.17 ± 0.122 , After = 5.659 ± 0.242 (unpaired t test, $t = 4.584$, $p = 0.0018$), China; Before = 3.718 ± 0.147 , After = 4.292 ± 0.241 (unpaired t test, $t = 2.739$, $p = 0.0255$), Korea; Before = 4.010 ± 0.218 , After = 5.647 ± 0.271 (unpaired t test, $t = 4.713$, $p = 0.0015$), Before; Orange (2012–2016), After; Blue (2017–2021), Mean $\pm$ SEM, * $p < 0.05$, ** $p < 0.01$.

Discussion

The results of this study highlight the significant impact that sex/gender-specific funding policies and journal editorial standards have on the integration of sex/gender considerations in biomedical research publications. The analysis demonstrates that countries with progressive funding policies, such as the United States, Canada, and the United Kingdom, have seen a marked increase in the quantity and quality of sex/gender-sensitive research outputs.

Role of sex/gender-sensitive policies and journal editorial standards

This study found that sex/gender-specific funding policies in the United States (through NIH's "Sex as a Biological Variable" [SABV] policy)^{35,50}, Canada (via CIHR's "Gender-Based Analysis Plus" [GBA+])^{37,42}, and the United Kingdom⁴⁰ (under UKRI's gender dimension policy) have led to an increase in both research grants and publications addressing sex/gender differences. However, in countries like Japan and China, where such policies are either less developed or absent, the corresponding research outputs have lagged. This suggests that funding alone is not sufficient; sustained policy advocacy and the creation of robust guidelines are necessary to cultivate a sex/gender-sensitive research environment.

Similarly, leading scientific and biomedical journals, including Nature^{52,53}, Cell Press⁵⁴, and the Lancet^{55,56}, have played a crucial role in encouraging research that incorporates a sex/gender perspective around 2017 and provided corresponding guidance. This initiative has spurred a significant increase in the quality and quantity of research publications that address sex/gender characteristics globally. This demonstrates the power of journals in influencing research practices globally, beyond the scope of national policies.

Differential impact by country

The interrupted time-series (ITS) analysis showed that the introduction of sex/gender-sensitive funding policies in the United States (2016), Canada (2017), and the United Kingdom (2018) had varying impacts. The United States experienced an immediate and sustained increase in both grant funding and publication output, particularly following the NIH's SABV policy. In contrast, Canada and the United Kingdom saw more gradual effects, with significant improvements only emerging over time. These differences highlight the importance of context-specific implementation strategies as cultural, institutional, and structural factors may influence how quickly such policies yield results.

Japan, China, and South Korea were not included in certain parts of our analysis due to the absence of formal sex/gender-sensitive funding policies, which limits our ability to fully assess the impact of such policies on research outcomes. However, we observed that the growth of sex/gender-inclusive research has been slower in countries without these formal policies. This suggests that the level of emphasis on 'sex/gender-inclusive research' policies can significantly influence the distribution of research funding and the pace of improvement in research outcomes. As these countries develop and implement such policies, future studies should monitor their effects on research outcomes to provide a more comprehensive understanding of global trends.

Research field variations

^{57,58} Integrated-sex/gender research is important in all research, but it is especially essential in some areas. These are the fields in which scientific evidence of sex differences exists, but the mechanisms are not yet understood. To better understand the trends in sex/gender-sensitive research, we selected four fields in the biomedical sciences that have a high proportion of sex/gender-sensitive research (psychology, biology, biomedical and clinical science, and health science) and analyzed the relationship between research grants and research outcomes in the USA and Canada, where research support policies are more visible. Overall, we found that psychology is the field with the highest level of integrated-sex/gender research support and outcomes. This prominence is likely driven by the significant sex/gender disparities observed in mental health conditions. For example, women are reported to be twice as likely to develop Alzheimer's disease as men^{17,59,60}, and autism spectrum disorders are more than four times more prevalent in men^{27,61–64}. In contrast, fields like biomedical and clinical sciences showed more modest increases in research output, despite significant funding support. These differences show that there is a great need for integrated-sex/gender research to uncover the underlying causes, which is why the number of sex/gender differences studies continues to grow. There is a reciprocal relationship between the expansion of research and awareness of sex/gender differences. Increased awareness leads to increased research, which in turn leads to greater understanding and awareness of sex/gender disparities. This dynamic cycle not only advances the field, but also promotes continued support and expansion of awareness of sex/gender differences in research. These findings suggest that while funding policies are essential, the nature of the research field and its scientific focus on sex/gender issues also play a critical role in shaping outcomes.

Limitations and implications for future research

There are several limitations to this study that must be acknowledged. First, the analysis was limited to English-language publications, which may have excluded important research from non-English-speaking countries, including Korea. This exclusion could result in an underestimation of sex/gender-integrated research in regions such as non-English-speaking countries. Future research should aim to include non-English-language studies to gain a more comprehensive view of global research trends. Second, due to data constraints, we were unable to analyze sex/gender-integrated research funding and outcomes in countries with emerging policies, such as Korea, where sex/gender-specific policies are still developing. Future studies should aim to include non-English-language publications and explore the impact of emerging sex/gender policies in regions such as Asia and Latin America.

Furthermore, while this study demonstrates a clear correlation between sex/gender-specific policies and research output, it did not control for potential confounding factors, such as overall research funding trends or

the influence of broader scientific advancements. Future research should seek to account for these variables and explore the long-term effects of sex/gender-sensitive policies on research quality and innovation.

Recommendations for policy and practice

Based on these findings, we recommend that governments and funding agencies worldwide adopt and expand policies that explicitly promote the integration of sex/gender dimensions in biomedical research. Such policies should include clear guidelines, training programs, and incentives for researchers to consider sex/gender in their studies. Additionally, leading scientific journals should continue to enforce editorial standards that require sex/gender analyses, as these have proven to be effective in promoting more inclusive research practices.

To ensure the continued growth of sex/gender-sensitive research, it is also critical that stakeholders invest in raising awareness of the importance of these considerations. Governments, academic institutions, and journals must collaborate to foster an environment where sex/gender dimensions are viewed as integral to scientific inquiry. Such research not only enriches scientific understanding, but also ensures more equitable health and technology solutions.

In addition to spreading awareness of sex/gender research, we propose a detailed study to ensure more effective and proactive support. A detailed keyword extraction and analysis of sex/gender characteristics in each field would reveal trends in more integrated-sex/gender research. This will allow us to identify areas where sex/gender research is concentrated and areas where active investment is needed, as well as the current status of sex/gender-sensitive research in various research field, beyond the biomedical field. This will allow us to discuss the trend of more inclusive sex/gender-sensitive research and how to promote it.

Data availability

Data are available upon request from the corresponding author.

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Authors' contributions

Kim H and Lee H were responsible for project administration and conceptualization; Kim H, Park J, and Ahn S took on methodology and data curation; Kim H drafted the original manuscript; and the review and editing were performed by Kim H, Park J, Ahn S, and Lee H.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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